

Amit Dutta

Reno, NV

amitdutta121@gmail.com | 775-219-7559

linkedin.com/in/amitdutta121 | github.com/Amitdutta121 | amitdutta.info

Visa: F-1 (OPT eligible) | Applied for NIW

SUMMARY

I am a Ph.D. researcher building AI-powered learning systems, including automated pipelines that generate and evaluate educational content using LLM judges and adaptive training systems that I have evaluated through user studies. I am also a Full-Stack Software Engineer with more than four years of experience building production mobile and web applications used by over 10,000 users. My engineering work includes ERP and bike-sharing systems, full-stack development with React, React Native, and Java Spring Boot, and CI/CD pipelines built with CircleCI, Fastlane, and CodePush.

EDUCATION

Ph.D. in Computer Science and Engineering

Aug 2023 – Present

University of Nevada, Reno, USA

Expected graduation: *Dec 2026*

Research focus: LLM evaluation, agentic AI, and adaptive machine learning.

MSc in Computer Science and Engineering

Jan 2021 – Feb 2023

BRAC University, Dhaka, Bangladesh

CGPA: 3.10

Thesis: Recurrent Proximal Policy Optimization for optimal block formation in Bitcoin blockchain networks.

TECHNICAL SKILLS

Software Engineering: TypeScript, FastAPI, Spring Boot, Node.js, React, PostgreSQL

LLM & Agentic AI: Large Language Models (LLMs), LLM Evaluation, LLM-as-a-Judge, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, LangSmith, MCP, MLflow, n8n

Programming & ML Systems: Python, C++, Java, SQL, Docker, Git, AWS

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Reinforcement Learning

EXPERIENCE

University of Nevada, Reno — Research Assistant

Aug 2023 – Present

- Developed an adaptive algorithm for personalized Python learning using Bayesian Knowledge Tracing and student performance.
- Developed an evaluation framework for LLM-generated university-level Python questions, combining deterministic validation, LLM-as-a-judge evaluation, rubric-based scoring, and professor feedback.
- Developed NAT-LLM, NAT, and RAFT for adaptive training in a Unity-based real-time ship simulator and evaluated the methods through human-subject studies.
- Conducted knowledge distillation experiments for visual question answering, comparing logit, attention, hidden-state, and hybrid distillation methods.
- Published first-author research at HCII 2025 and HCII 2026 on adaptive training systems.

Brainstation23 — Senior Software Engineer

Oct 2021 – Jul 2023

- Built and shipped production mobile and web applications using **React Native** and **React**.
- Designed and maintained CI/CD pipelines using **CircleCI**, **Fastlane**, and **CodePush** to automate testing, deployment, and over-the-air updates.
- Led and mentored multiple developers in building reusable components, authentication flows, payment integrations, and release pipelines, improving development speed and maintainability across projects.
- Developed an **AdaBoost** model to predict rental-vehicle demand and prioritize charging locations, achieving **98% accuracy**.

Sweetitech — Software Engineer

Oct 2020 – Nov 2021

- Delivered full-stack features for ERP, HR, and pharmaceutical applications using **React**, **React Native**, and **Java Spring Boot**.
- Developed REST APIs and **MySQL** backend services supporting thousands of users.

PROJECTS

- **Adaptive Python Tutor and LLM Evaluation System** Aug 2023 – Present [GitHub](#)
Universal Adaptive Trainer is a PhD research platform that turns a textbook into an adaptive Python course. I built the full pipeline: retrieval-augmented generation (RAG) of questions against a topic taxonomy, a deterministic eval that sandboxes and type-checks every generated question before a professor reviews it, and an LLM-as-judge layer whose prompts I align to professor feedback using GEPA. Students open a generated link where Bayesian Knowledge Tracing selects each next question from their performance and weak topics. Backend in FastAPI with structured LLM calls, frontend in Next.js and TypeScript.
- **Neuro-Adaptive Training with AI-Generated Feedback** Aug 2025 – Present [Publication](#)
Developed **NAT**, a domain-agnostic adaptive training algorithm that selects each student's next activity from cognitive load and performance, and **NAT-LLM**, an expert-guided LLM feedback system. Applied both in a Unity naval navigation simulator, where NAT significantly outperformed non-adaptive training. NAT-LLM's **few-shot** feedback is preferred by 73% of the participants.
- **Multimodal Knowledge Distillation for Visual Question Answering** Nov 2025 – Present
Developed a Multimodal Knowledge Distillation pipeline comparing logits, attention, hidden-state, and hybrid distillation strategies; identified the best-performing distillation approach on a custom dataset.
- **Thinker** Aug 2025 – Present [GitHub](#)
MyThinker is a planning agent that refines your plan as you chat. It keeps a persistent memory across sessions and uses a graph-based idea fan-out algorithm to branch and explore ideas. The model does the reasoning. The frontend is built with Next.js and TypeScript. The backend is built with FastAPI and serves a streaming API. LangGraph runs the agent graph, Pydantic defines the schemas, and SQLite stores the memory. MLflow is implemented for observability and eval generation.
- **Flour** Aug 2025 – Present [GitHub](#)
Flour is a Windows desktop app that turns speech into text in real time, fully on-device. I built the streaming audio pipeline with Silero VAD endpointing that feeds local ASR models (Whisper, Parakeet, Moonshine) run through ONNX, plus a local LLM via Ollama that cleans up the transcript with per-app context. I fine-tuned a speech-correction model to keep latency low. UI is implemented in React and TypeScript, history in SQLite.
- **Traffic Anomaly Detection System** Aug 2025 – Present
Designed an end-to-end **MLOps** pipeline for anomaly detection on traffic sensor data using **Python**, **MLflow**, and **DVC** for experiment tracking, artifact management, and reproducibility. Containerized model serving with **FastAPI** and **Docker**, and implemented **Prometheus** / **Grafana** monitoring for real-time metrics, observability, and performance tracking.
- **Sharebike** [Android](#) [iOS](#)
Built an end-to-end white-label bike-sharing mobile application using **React Native**, serving over **10,000 users**. Integrated **Firebase Authentication**, **Stripe** payments, **CircleCI** for automated deployments, and **CodePush** for over-the-air updates.
- **Sharebike Backend: Fleet Management Microservices** Oct 2021 – Jul 2023
Designed and built Sharebike's backend fleet-management services, using **Spring Boot** microservices for multi-region fleet management and event-driven, **Google Cloud Pub/Sub**-based ingestion for real-time device data processing across the fleet.
- **ROBB: Recurrent PPO for Blockchain Block Formation** [Publication](#)
Designed and evaluated a recurrent reinforcement learning approach for dynamically creating blocks in a blockchain network efficiently. The proposed algorithm balances waiting time against block utilization and was published in *IEEE Access*.

PUBLICATIONS

1. Dutta, A., & Louis, S. J. (2025). Evaluating Adaptive Training for Nautical Rules of the Road. In *International Conference on Human-Computer Interaction* (pp. 19–34). Springer, Cham.
2. Dutta, A., & Louis, S. J. (2026). Weakness Adaptation in Adaptive Training for Nautical Rules of the Road. In *International Conference on Human-Computer Interaction* (pp. 301–317). Springer, Cham.
3. Dutta, A., Rafin, N. I., Dewan, M. A. A., & Alam, M. G. R. (2024). ROBB: Recurrent Proximal Policy Optimization reinforcement learning for optimal block formation in Bitcoin blockchain networks. *IEEE Access*, 12, 31287–31311.

ACHIEVEMENTS

- 1st place in Datathon, Dhaka International University, Bangladesh